Tahmidul Bin Ferdous

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Education

BSc in Computer Science & Engineering *2022 – 2026*

International Islamic University Chittagong (IIUC), Chittagong, Bangladesh

Undergraduate Thesis: "Detecting Data Leakage in Lung Cancer CT Classification: A Patient-Level Evaluation of CNN, EfficientNet, and Vision Transformer Models."

GPA: 3.31 / 4.00

Technical Skills

- **Programming Languages:** Python (Primary, 250+ logged experiments), C++, SQL, Bash/Shell, LaTeX.
- **Deep Learning Frameworks:** PyTorch (Advanced), JAX/Flax, Optax, torchvision, timm, TensorFlow/Keras, Hugging Face.
- **Computer Vision & ML Libraries:** OpenCV, MediaPipe, Ultralytics YOLOv11, scikit-learn, Albumentations, NumPy, Pandas, Matplotlib.
- **Systems & Infrastructure:** Distributed Data Parallel (`torchrun`, multi-GPU DDP), Google Cloud TPU v5e-8 (8-device `jax.pmap`), Mixed Precision (AMP / BF16), Weights & Biases (250+ runs), Git, Docker, Linux/Ubuntu.
- **Models Built from Scratch:** Mamba-Transformer Hybrid (**DSMaT-Pico/Nano**), Multi-Teacher Distillation, Class-Conditional Rectified Flow (Euler ODE), Conditional DDPM, Autoregressive GPT-1, WGAN-GP, DCGAN.

Research & Professional Experience

EliteLab *Remote*

Student Researcher *2026 – Present*

- Conducting empirical research on benchmark contamination, shortcut learning, and evaluation validity across vision and generative models.
- Performing LLM-assisted post-mortem code and pipeline audits across multi-GPU PyTorch and multi-device JAX workloads to verify data partition isolation and reproductive fidelity.

Outlier AI *Remote*

AI Data Quality & Alignment Specialist *November 2024 – February 2025*

- Audited and curated large-scale human preference and reasoning datasets utilized for Reinforcement Learning from Human Feedback (RLHF) and alignment.
- Refined multi-turn annotation guidelines, directly mitigating prompt ambiguity and improving inter-annotator reliability across reasoning benchmarks.

Research & Engineering Projects

Data Leakage Audit in Lung Cancer CT Classification (Undergraduate Thesis) *Sep 2025 – Jun 2026*

PyTorch, timm, CCAL, EfficientNet B0–B4, ViT-B/16, Grad-CAM, W&B (188 runs) | *Code* | *W&B Logs*

- Derived via the hypergeometric distribution that standard 80/20 slice-level splits on the IQ-OTH/NCCD lung CT dataset (1,097 slices, 110 patients) leak **83.5% of minority Benign patients** across train and test sets.
- Audited 11 architectures across 4 model families (CCAL dual-CNN + MHA, EfficientNet B0–B4, ViT-B/16, ViT-B/32, ViT-L/16, and a PLOS ONE 4-block CNN) under paired leaky 80/20 vs. sequential case-order 70/20/10 splits, showing test accuracy drops of **6.07 to 18.28 percentage points** (CCAL 99.55% $\rightarrow$ 85.71%; EfficientNet-B0 98.64% $\rightarrow$ 91.07%) and Benign F1 collapse from **0.66–0.98 down to 0.17–0.62**.
- Conducted Grad-CAM saliency and MHA attention-weight diagnostics plus an AI-assisted post-mortem code audit, uncovering peripheral chest-wall/scanner-table shortcut learning and ASCII filename boundary-cutting artifacts.

DSMaT (MambaVision-Hybrid) & Multi-Teacher Knowledge Distillation *Jan 2026 – Mar 2026*

PyTorch, 2D Selective SSM (Mamba), Windowed Self-Attention, KL Distillation, Weights & Biases | *Code*

- Engineered **DSMaT-Pico (0.84M params)** and **DSMaT-Nano (2.04M params)** from scratch by hybridizing Depthwise Separable Convolutions (Stages 1–2) with symmetric 2D Mamba Selective Scans and Windowed Self-Attention (`MMSS`, Stages 3–4), reaching **97.58%** stratified test accuracy at **4.8×–17× smaller size** than EfficientNet-B0 and CCAL baselines.
- Built a Multi-Teacher Knowledge Distillation pipeline ($T = 4.0, \alpha = 0.3$) distilling an ensemble of `EfficientNet-B0`, `DenseNet-121`, and `ConvNeXt-Tiny` (38.8M combined) into a **0.35M-parameter** `ShuffleNetV2-0.5x` **student**, outperforming the best individual teacher on the strict sequential split (**88.39% vs. 87.50% accuracy**, Benign F1 0.3636 $\rightarrow$ 0.5833, 0.9646 AUC) at **20.2× parameter compression**.

Conditional Rectified Flow & DDPM for Minority-Class CT Synthesis (GPU DDP $\rightarrow$ TPU v5e-8) *Feb 2026 – Mar 2026*

PyTorch DDP, JAX/Flax (`pmap`), Google Cloud TPU v5e-8, 2× Tesla T4, W&B (23 runs, 36.3 GPU/TPU-hrs) | *RF Code* | *DDPM Code*

- Implemented a 35.72M Class-Conditional DDPM ($T = 1000$) and reformulated it into straight-line **Rectified Flow** ($x_t = (1-t)x_0 + t\varepsilon$, Logit-Normal timestep sampling, 50-step Euler ODE solver, **20× faster sampling**); resolved a Classifier-Free Guidance null-token index

collision (`null_class=3`) to boost downstream `DenseNet-121` sequential-split accuracy to **88.39%** and **0.6667 Benign F1** (surpassing classical augmentation).

- Scaled the Conditional U-Net to **73.46M parameters at native** 224×224 **resolution**: after diagnosing a `torch.compile` CUDA Graph memory-pool OOM on $2\times$ Tesla T4 GPUs (33.5s/epoch over 873 epochs), ported the full training and fused-CFG pipeline to **native JAX/Flax (`jax.pmap`) on an 8-core Google Cloud TPU v5e-8**, achieving a **13.4$\times$ speedup (2.50s/epoch)** and completing 3,000 epochs in 2.1 hours (`best_loss = 0.0923`).

BanglaPoemGPT — 10M-Parameter Autoregressive Transformer from Scratch *March 2025*

Python, PyTorch, Kaggle P100 (4 runs, 2.0 GPU-hours) | *Code*

- Engineered a GPT-1 architecture from first mathematical principles without high-level library wrappers: multi-head causal self-attention, learned positional embeddings, custom tokenizer, and autoregressive generation loop; trained on a 3M-token Bengali poetic corpus to a validation cross-entropy loss of **1.693**.

Edge Vision: YOLOv11 Campus Security & Facial Biometric Systems *Oct 2024 – Nov 2024*

YOLOv11, OpenCV, FaceNet, PyTorch | *Code, Kaggle* | *Demos: ID Card Detection, Facial Attendance*

- Curated, annotated, and trained custom YOLOv11-s/m models for campus ID compliance monitoring (**mAP50 = 0.82**), and built an edge facial recognition pipeline using SSD detectors and FaceNet 128-d cosine similarity matching.

Generative Adversarial Networks: Bengali Digits & Faces *Nov 2024*

PyTorch, DCGAN, WGAN-GP, Weights & Biases (42 runs) | *Code (Digits), Code (Faces)*

- Implemented DCGAN and WGAN-GP (gradient penalty, critic loop) from scratch to generate handwritten Bengali digits and face images, comparing training stability across adversarial objectives.

Interactive Assistive & Real-Time Vision: Sketch AI & Face Mouse *Aug 2024 – Nov 2024*

MediaPipe, OpenCV, PyAutoGUI, Python | *Demos: Sketch AI, Face Mouse*

- Built a real-time webcam air-canvas driven by fingertip landmarks (**Sketch AI**) and a hands-free cursor controller for users with motor disabilities where nose landmarks steer the cursor and eye blinks trigger clicks (**Face Mouse**).

Certifications

- **CS231n: Deep Learning for Computer Vision, DeepLearning.AI Specializations (Coursera):** Neural Networks & Deep Learning, Improving Deep Neural Networks (Tuning, Regularization, Optimization), Convolutional Neural Networks, Sequence Models, Data Scientist Machine Learning (Codecademy).

Academic References

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